OS JACKFRUIT PROBLEM SCREENSHOT OUTPUTS

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```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine start alpha ./rootfs-alpha /cpu_hog --nice 10
[sudo] password for vboxusercdn:
started container 'alpha' pid=7579 soft=40MiB hard=64MiB
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine start beta  ./rootfs-beta  /cpu_hog --nice -5
started container 'beta' pid=7588 soft=40MiB hard=64MiB
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine ps
  alpha            pid=7579    running            started=16:38:34  soft=40MiB  hard=64MiB
  beta             pid=7588    running            started=16:38:46  soft=40MiB  hard=64MiB
```

1. **Multi-container Supervision**
   The status output confirms that the supervisor is managing two concurrent containers, 'alpha' and 'beta', each assigned a unique process ID.

```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine ps
  alpha            pid=7579    running            started=16:38:34  soft=40MiB  hard=64MiB
  beta             pid=7588    running            started=16:38:46  soft=40MiB  hard=64MiB
```

2. **Metadata Tracking**
   The output of sudo ./engine ps displays the supervisor's tracked metadata for each container, including PIDs, uptime, and memory configurations.

```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine logs alpha
sh: /cpu_hog: not found
sh: /cpu_hog: not found
cpu_hog: spinning...
cpu_hog: tick 268435456
cpu_hog: tick 536870912
cpu_hog: tick 805306368
cpu_hog: tick 1073741824
cpu_hog: tick 1342177280
cpu_hog: tick 1610612736
cpu_hog: tick 1879048192
cpu_hog: tick 2147483648
cpu_hog: tick 2415919104
cpu_hog: tick 2684354560
cpu_hog: tick 2952790016
cpu_hog: tick 3221225472
cpu_hog: tick 3489660928
cpu_hog: tick 3758096384
cpu_hog: tick 4026531840
```

3. **Bounded-buffer Logging**
   The output of sudo ./engine logs alpha verifies that the logging pipeline successfully captured and stored the cpu_hog ticks through the bounded buffer.

```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine stop alpha
sent SIGTERM to container 'alpha' (pid=7579)
```

4. **CLI and IPC**

   The sudo ./engine stop alpha command demonstrates the end-user request being sent via CLI, triggering the supervisor to respond with a SIGTERM via IPC.

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```
usercdn@UbuntuVM:~/Documents/OS_mini$ sudo cp -a rootfs-base rootfs-memtest
usercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine start memtest ./rootfs-memtest "/memory_hog 16
--soft-mib 32 --hard-mib 64
ted container 'memtest' pid=7965 soft=32MiB hard=64MiB
usercdn@UbuntuVM:~/Documents/OS_mini$ sudo dmesg | grep container_monitor
98.641732] [container_monitor] Loaded.
46.594791] [container_monitor] registered alpha pid=6832 soft=0 hard=0
46.649096] [container_monitor] unregistered pid=6832
66.416793] [container_monitor] registered alpha pid=6884 soft=0 hard=0
66.448747] [container_monitor] unregistered pid=6884
97.170151] [container_monitor] registered alpha pid=7093 soft=0 hard=0
97.417790] [container_monitor] alpha (pid 7093) gone, removing
58.103084] [container_monitor] registered beta pid=7106 soft=0 hard=0
58.259849] [container_monitor] beta (pid 7106) gone, removing
67.530908] [container_monitor] registered alpha pid=7579 soft=41943040 hard=67108864
68.291567] [container_monitor] alpha (pid 7579) gone, removing
79.747246] [container_monitor] registered beta pid=7588 soft=41943040 hard=67108864
80.456188] [container_monitor] beta (pid 7588) gone, removing
17.053749] [container_monitor] registered memtest pid=7965 soft=33554432 hard=67108864
18.332133] [container_monitor] memtest (pid 7965) gone, removing
```

5. Soft-limit Warning

The dmesg logs show the container_monitor kernel module registering the memory thresholds and tracking the RSS usage for the 'memtest' container.

```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ sudo ./engine supervisor ./rootfs-base
[sudo] password for vboxusercdn:
[engine] supervisor ready  socket=/tmp/mini_runtime.sock
[engine] container 'beta' pid=7588 reaped (state=hard_limit_killed)
[engine] container 'alpha' pid=7579 reaped (state=stopped)
```

6. Hard-limit Enforcement

The supervisor log confirms that 'beta' was reaped with the state hard_limit_killed after the kernel module detected it exceeded the 64MiB hard limit.

```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ ps aux | grep cpu_hog
root        7093 83.9  0.0   1004   584 pts/3    RN+  16:10  27:11 /cpu_hog
root        7106 98.1  0.0   1004   584 pts/3    R<+  16:11  30:46 /cpu_hog
root        7579  6.5  0.0   1004   584 pts/4    RN+  16:38   0:17 /cpu_hog
root        7588 90.9  0.0   1004   584 pts/4    R<+  16:38   3:57 /cpu_hog
vboxuse+    7846  0.0  0.0   9144  2248 pts/2    S+   16:43   0:00 grep --color=auto cpu_hog
```

7. Scheduling Experiment

The ps aux output compares the two cpu_hog processes, showing the high-priority container (nice -5) receiving significantly more CPU time than the low-priority one.

```
vboxusercdn@UbuntuVM:~/Documents/OS_mini$ ps aux | grep defunct
vboxuse+    8126 16.6  0.0   9144  1984 pts/2    S+   16:52   0:00 grep --color=auto defunct
```

8. Clean Teardown

The ps aux | grep defunct command confirms that all child processes were properly reaped by the supervisor, leaving no zombie processes in the system.